सत्यं शिवं सुन्दरम्

Faculty of Technology & Engineering
The Maharaja Sayajirao University of Baroda

PROJECT REPORT

FS BE-III MICROPROCESSOR AND MICROCONTROLLER

Interface 8-digit 7 Seg LED display and 8 keys. Display 8-digit message in 8 different ways.

DONE BY:

Swapnil Varma (8024058406)

ACADEMIC YEAR: 2025-26

AIM

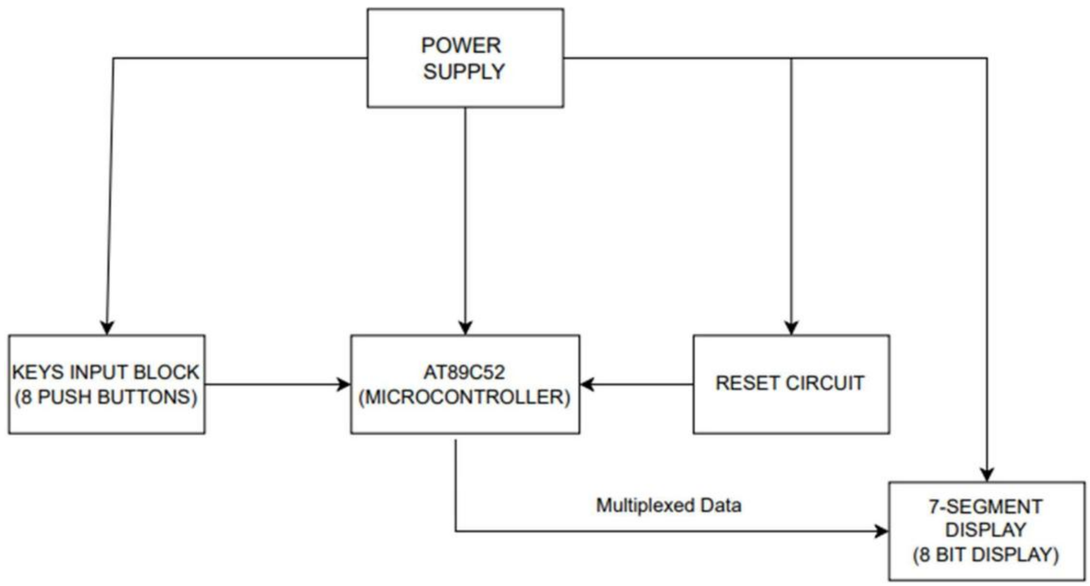
Interface 8-digit 7 Seg LED display and 8 keys. Display 8-digit message in 8 different ways as follows as per key pressed:

- (key-1) Steady message
- (key-2) Blinking message
- (key-3) Rolling through left
- (key-4) Rolling through right
- (key-5) Odd no LED's blinking others steady
- (key-6) Even no LED's blinking other's steady
- (key-7) Rolling half message from outside to inside
- (key-8) Rolling half message from inside to outside

LIST OF COMPONENTS

Sr. No.	Components	Quantity
1	AT89C52 40 Pin IC Base	1
2	Push Button (5mm)	8
3	7-Seg LED Display(Common Anode)	8
4	USB Connector	1
5	1000 µF/35V Capacitor	1
6	33 pF/25V Capacitor	2
7	470 µF/25V Capacitor	1
8	10 µF/35V Capacitor	1
9	104 Capacitor	3
10	330E Resistors	9
11	10k Resistors	8
12	4k7 Resistors	1
13	IC 7805 (Voltage Regulator 5V)	1
14	BC558 (PNP) Transistors	8
15	IN4148 Diode (Zener)	1
16	IN4007 Diode	5
17	LED (5mm)	1
18	12MHz Crystal	1
19	10k, 9 Pin R-Pack	1

BLOCK DIAGRAM



BLOCK DESCRIPTIONS

1. Power Supply

- Provides the required +5V regulated DC supply to the microcontroller (AT89C52), display, and input circuit.
- Typically consists of:
  - Step-down transformer (230V AC → 12V AC),
  - Bridge rectifier (AC → DC),
  - Filter capacitor (to remove ripples),
  - Voltage regulator (7805) to maintain +5V stable output.
- Ensures stable operation of logic circuits and LEDs.

2. Reset Circuit

- Provides a power-on reset to the AT89C52 microcontroller.
- Consists of a resistor-capacitor (RC) network and a push-button.
- On power-up, it ensures the microcontroller starts execution from the reset vector (0000H).
- Manual reset option allows the user to restart the program anytime.

3. Keys Input Block (8 Push Buttons)

- 8 push buttons are connected to one port of the microcontroller.
- Each key corresponds to one display function:
  - Key-1: Steady message
  - Key-2: Blinking message
  - Key-3: Rolling left
  - Key-4: Rolling right
  - Key-5: Odd digits blink, even steady
  - Key-6: Even digits blink, odd steady
  - Key-7: Half message rolls from outside → inside
  - Key-8: Half message rolls from inside → outside
- The microcontroller continuously scans the keys (polling or interrupt method) and selects the display mode accordingly.

4. AT89C52 Microcontroller

- Acts as the control unit of the system.
- Main tasks:
  - Reads the pressed key from the input block.
  - Fetches the stored 8-digit message (in code memory).
  - Generates multiplexed signals for the 7-segment display.
  - Applies the selected effect (steady, blinking, rolling, etc.).
- Features of AT89C52 relevant here:
  - 4 I/O ports → can handle display multiplexing and input.
  - Internal timers used for delay generation (blinking, rolling).
  - Fast control logic to refresh display and maintain persistence of vision.

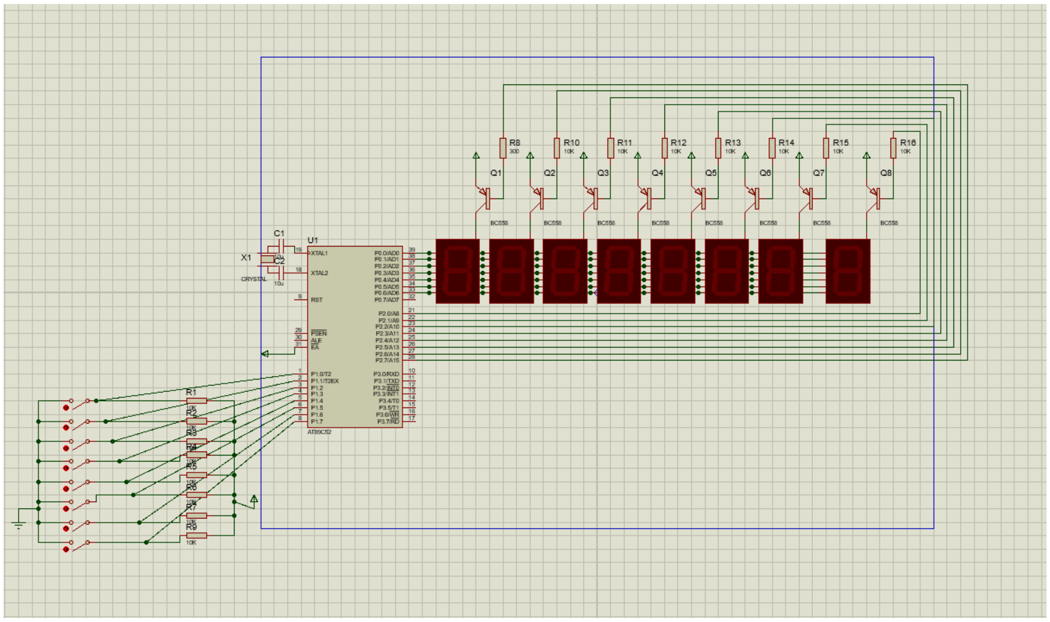
5. 7-Segment Display (8-Digit, Multiplexed)

- Consists of 8 individual 7-segment LED modules arranged side by side.
- Displays 8-digit alphanumeric message.
- Multiplexing is used:
  - Only one digit glows at a time, but switching is fast.
  - To human eyes, it appears as if all digits glow simultaneously.
- Controlled by:
  - Segment lines (a–g, DP) connected commonly.
  - Digit enable lines (through transistors or drivers like ULN2803).
- The microcontroller updates display continuously to show effects like blinking or rolling.

Working (Overall Flow)

1. User presses a key → microcontroller detects it.
2. Microcontroller selects the corresponding display mode.
3. Message is stored in program memory → sent to display driver logic.
4. Using multiplexing technique, display is refreshed at high speed.
5. Depending on the mode:
   - Steady → fixed display.
   - Blinking → alternates between ON and OFF.
   - Rolling → shifts message left or right.
   - Odd/Even blinking → selective digits blink.
   - Outside → inside roll → message parts shift toward center.
   - Inside → outside roll → message parts shift outward.

SCHEMATIC DIAGRAM



CONNECTIONS:

8051 consists of 4 PORTS out of which:

- PORT 0 is connected to the Pins of 7 Segment LEDs in Common Anode Configuration via a 330Ω resistors on the Data Bus.
- PORT 2 acts as Segment Drive consisting of 8 PNP transistors whose Collectors are connected to Common Anode of the 7 Segment LEDs, Emitters are connected to a common Ground and Base are connected to 8 different Pins in Port 2 via 10k registers.
- PORT 3 is free for future use.
- PORT 1 contains 8 individual keys, each used for displaying different message:
    - Key 1-Steady message
    - Key 2-Blinking message
    - Key 3-Rolling through right
    - Key 4-Rolling through left
    - Key 5-Even no LEDs blinking other's steady
    - Key 6-Odd LEDs blinking other's steady
    - Key 7-Rolling half message from outside to inside

- Key 8-Rolling half message from inside to outside. Keys are connected in Pull Down configuration given 5 V common supply connected to 10k register pack.

- Crystal Oscillator is connected to Pins XTAL 1 and XTAL 2 with two 33pF capacitors in parallel representing the crystal circuit.
- Reset circuit is connected to Pin RST in Power ON Reset configuration.
- Input Supply is given through a 12V adapter which passes through a bridge rectifier and Regulator IC 7805 to get a 5V constant supply for the 8051 Microcontroller.

PCB LAYOUT :